

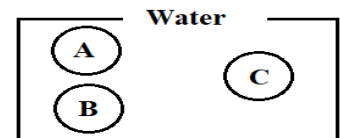
Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ A/E	First Test - Modeule B	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **8 Marks**

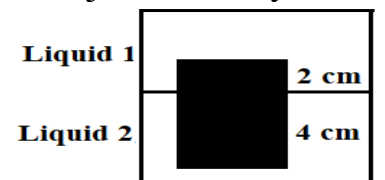
1) The acceleration is zero for floating object		
2) The electric field is a region around a charge		
3) Partially immersed is only related to Archimedes Principle		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) The direction of E lines of a charge q depends on the:				
a) Value of q	b) Direction of q	c) Magnitude of q	d) Sign of q	e) None of these
2) The electric force is given by:				
a) $K_e q/r^2$	b) $k_e q_1 q_2/r$	c) $k_e q_1 q_2/r^2$	d) $k_e q/r$	e) None of these
3) 13 protons and 6 electrons. The total charge is:				
a) 13 e	b) 7 e	c) - 7 e	d) - 13 e	e) None of these
4) From the figure shown below. The pressures				



a) $P_C < P_A$	b) $P_A = P_C$	c) $P_B > P_C$	d) $P_A > P_B$	e) None of these
5) A cube of x each side is placed in $\vec{E} = \alpha \hat{i} - \beta \hat{j} - \gamma \hat{k}$. The electric flux in yz and xy planes is:				
a) $-(\gamma + \beta) x^2$	b) $(\alpha - \gamma) x^2$	c) $(\alpha - \beta) x^2$	d) $-(\alpha - \beta) x$	e) None of these
6) An object of ρ suspended between liquid 1 of ρ_1 and liquid 2 of ρ_2 . The object's density is:				



a) $\rho = 2 \rho_1 + 4 \rho_2$	b) $\rho = 4 \rho_1 + 2 \rho_2$	c) $\rho = \frac{4}{6} \rho_1 + \frac{2}{6} \rho_2$	d) $\rho = \frac{2}{6} \rho_1 + \frac{4}{6} \rho_2$	e) None of these
7) If a ratio of object to fluid densities is 1.8. Then the fully immersed object is:				
a) Sinking	b) Floating	c) Moving up	d) At rest	e) None of these

Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

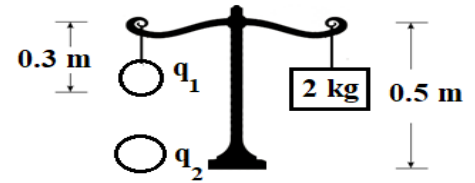
20 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31}$ kg	$K_e = 9 \times 10^9$ N.m ² /C ²	Gravity: $g = 10$ m/s ²
Proton: $m = 1.67 \times 10^{-27}$ kg	$G = 6.67 \times 10^{-11}$ N.m ² /kg ²	Water: $\rho_f = 1000$ kg/m ³

1) 500 N object floats in water of $\rho_f = 1000$ kg/m³, the object is immersed by 7 cm and 5 cm over the water. The density of the object in kg/m³ is:

a) 0.086	b) 583.3	c) 0.119	d) 1400	e) None of these
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2) 2 kg at one side of a balance scale and a charge $q_1 = 10 \text{ nC}$ at the other as shown. The value of the charge q_2 for equilibrium is:

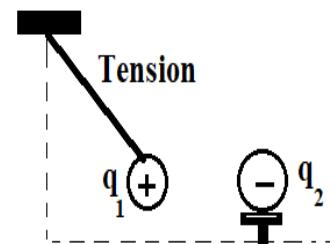


a) - 0.0089 C	b) 0.01778 C	c) - 0.01481 C	d) 0.0296 C	e) None of these
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3) $q_1 = - 2 \text{ nC}$ is placed at $(- 0.3 \text{ m}, 0)$, $q_2 = - 2 \text{ nC}$ is placed at $(0 , - 0.3 \text{ m})$. The electric field in N/C at the origin is:

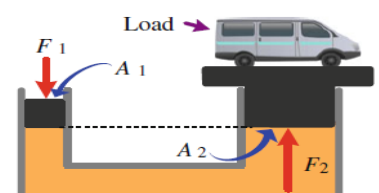
a) 424.3 at 315°	b) 282.8 at 225°	c) 424.3 at 225°	d) 282.8 at 45°	e) None of these
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4) 100 g and $q_1 = 2 \text{ }\mu\text{C}$ charge hung by a rope. A charge of $q_2 = - 9 \text{ }\mu\text{C}$ attracts q_1 charge. Then the rope inclined and the distance between the charges is 20 cm. The tension in the rope is:



a) 4.2 N	b) 120.6	c) 2.9 N	d) 3.7 N	e) None of these
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5) In Pascal principle: The piston areas are 25 cm^2 and 750 cm^2 respectively, the weight on the input piston is 10000 N. The weight that the output piston handle is:



a) 3000 N	b) 33333.3 N	c) 333.3 N	d) 300000 N	e) None of these
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